

BREAKHOOK · DESIGN DECISION RECORD

BreakHook prop holder stand-off accepted

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DOCUMENT CONTROL

Document ID	BHK-DDR-004	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breakhook		
Source	docs/decisions/0004-prop-holder-distance.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Amish's round-2 decision 5A on the prop holder's distance recorded

0004: Prop holder at 2.9 m for placement only

- **Date:** 2026-10-03
- **Status:** decided by Amish, 2026-10-03 (round 2): "i agree with all the 46 recommendations you provided. please proceed." For BreakHook this is decision 5A, the open decision in BHK-DEC-001 v0.2.

■ Context

The fork prop (BHK-DDR-003) works only near the pole's mid-length. There its holder stands about 2.9 m from the wall for the 25 s of placement, 1.1 m nearer than R1's 4 m front-hand distance. The rear handler is 5.06 m away. The firebreak is opened two dwellings ahead of the fire (BHK-DDR-001, decision 10), so the dwelling being hooked is not yet burning.

■ Options considered

Option	What it means
A (chosen)	Accept 2.9 m for the prop holder during placement only; restate R1's distance as applying to the pole hands; the holder withdraws with the pole before the haul
B	Keep 4 m for everyone and move the prop back (droop 0.72 m, no better than by hand)
C	Keep 4 m and set the prop, then let it stand alone in the fork while its holder steps back (untested)

■ Decision

Option A. No change to parts. R1 is restated: "Places the hook at a height of 3 m (10 ft) with the pole hands (front hand) 4 m (13 ft) from the wall; the prop holder may stand 2.9 m from the wall during placement only (about 25 s) and withdraws with the pole before the haul". The reach trial at TRL 4 confirms the distance, the time at 2.9 m and the withdrawal.

Table 1. Effects.

Item	Result
R1	Met on paper as restated: droop about 0.27 m, rear hand 5.06 m, prop holder 2.9 m for about 25 s (BHK-CAL-001 [F19], [F24]); confirmed at TRL 4
Other requirements	None changed; pole and hook 7.82 kg
Wording changed	Requirements v0.4 (R1), build plan (reach and drill checks), register v0.3
Cost	Unchanged. Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 1,170 (USD 830 under the target). <code>budget_usd</code> unchanged

■ Safety

Safety: BreakHook pulls down a structure with a rope: haulers stand at least 10 m from the wall and at least 1.5 times the structure's height away. The prop holder is the nearest person to the wall, at 2.9 m, only while the hook is set and the structure is not burning; the holder withdraws with the pole before anyone hauls, and nobody is inside the fall zone at the haul. Overhead lines stay at least 3 m clear. It is an open engineering reference, not certified equipment.